

Weight EMA after separating training duration from learning-rate policy in a small diffusion model

allagma

9 October 2026

Abstract

Exponential moving averages (EMA) of neural-network weights can improve a model during training, but their relative benefit depends on the trajectory being averaged. We present a new manuscript revision of a completed, bounded two-dimensional diffusion study that crossed terminal duration with the full optimizer learning-rate policy. The retained experiment compared raw weights with EMA decays of 0.99 and 0.999 on two synthetic datasets, using 4 independent paired seeds per dataset. It completed 32 evaluation cells and retained 96 model states. EMA-minus-raw SW1 was negative in 14 of 16 cell-level means, but only 2 unadjusted 95% seed-level intervals were wholly below zero. The cosine-minus-constant contrast in the paired EMA effect was positive in all 8 comparisons; 2 corresponding intervals were wholly above zero. All 8 duration-contrast intervals and all 4 interaction intervals included zero. These results describe the selected model, datasets and budgets. They agree qualitatively with prior work connecting averaging and annealing, but do not identify an endpoint-only mechanism, an optimal decay, or a general EMA benefit. The revision adds a critical literature map and portable paper sources without rerunning or altering the historical experiment.

1 Introduction

Weight averaging offers a way to evaluate a training trajectory through more than its final iterate. Diffusion-model implementations often retain an EMA of the trained parameters, including the original DDPM work [2]. This practice does not establish that every averaging rule improves every endpoint. The optimizer, learning-rate path, averaging window and evaluation procedure all affect the comparison.

A smaller difference between averaged and raw weights also has two possible interpretations: the average may have become worse, or the raw model may have caught up. A useful comparison therefore reports both the paired relative effect and the absolute quality of each model. The distinction matters when comparing constant learning rates with a schedule that approaches zero.

The retained study asks how the relative benefit of fixed-decay EMA changes across terminal training duration and the complete learning-rate policy in a small DDPM. It separates complete cosine schedules at two durations from a constant-rate trajectory evaluated at the same update counts. The study uses simple two-dimensional distributions so that paired model states, generated samples and inputs can be retained and checked within a finite local budget. It does not seek a new state of the art or establish a general explanation of averaging. In particular, the relationship between averaging and annealing is already directly examined by Ajroldi et al. [1].

This paper is a publication revision of completed Allagma run r07. The literature search described

here occurred after that experiment. Its role is to correctly situate and qualify the retained findings, not to retroactively change the protocol or claim that these sources determined its original freeze.

2 Related work

DDPM trains a noise-prediction model and generates samples through a learned reverse process [2]. Ho et al. report parameter EMA with a fixed decay without a decay sweep in their experimental details. Nichol and Dhariwal introduce a cosine schedule for the forward diffusion noise [7]. That schedule concerns corruption over diffusion time. It is distinct from the optimizer learning-rate policy over training updates, which is the intervention in the retained experiment.

Stochastic Weight Averaging (SWA) combines selected SGD iterates under constant or cyclical learning rates [3]. Its averaging rule differs from a fixed exponential decay applied after every AdamW update. SWA is a relevant alternative that this experiment did not test, so the results cannot rank EMA against it.

Morales-Brotons et al. study EMA dynamics and report strong intermediate performance before full learning-rate reduction in their classification experiments [6]. Their validation-based early stopping, SGD trajectories and batch-normalization choices differ from the fixed endpoints here. Ajroldi et al. subsequently compare averaging across a wider optimization benchmark and examine no, partial and full annealing [1]. Their results show that averaging gains can diminish as the baseline is fully annealed and that combining the two can give the best final performance in the settings tested. This is direct prior evidence for the qualitative relationship considered in this paper.

For diffusion models, Karras et al. show that preferred averaging profiles depend on architecture and training time [4]. Their power-function and post-hoc averaging procedures, image-scale models and FID evaluation differ from the two fixed decays and SW1 metric here. Their findings support caution about transferring a preferred averaging window across designs.

The literature also contains contrary ordinary-EMA outcomes. Li et al. report video-prediction and regression cells where ordinary EMA performs worse than the basic model, while proposing a switching procedure that feeds averaged parameters back into training [5]. That intervention changes the training trajectory. Its reported gains and the unhelpful bootstrapping experiment in Morales-Brotons et al. are not matched comparisons. The present study evaluates averages of the same raw trajectories and cannot resolve that protocol-dependent difference.

3 Methods

3.1 Retained design and implementation

The experiment crossed two datasets (two moons and an eight-component Gaussian mixture, GMM8), two terminal durations (5,000 and 10,000 optimizer updates), and two learning-rate policies. The constant policy used $\eta_0 = 0.0003$. For duration D , update k under cosine used

$$\eta_k(D) = \frac{\eta_0}{2} \left[1 + \cos\left(\frac{\pi(k-1)}{D}\right) \right]. \quad (1)$$

The final applied rate is slightly above zero, while the scheduled rate after D updates is zero. The policies differ over the entire training path; the experiment does not isolate an effect caused only by the terminal rate.

Each constant-rate trajectory supplies both checkpoints. Cosine-5k and cosine-10k are separate trajectories initialized with the same weights and matched training-input prefixes. The 32 evaluation cells therefore come from 24 trajectories totaling 200000 optimizer updates, rather than one independent trajectory per checkpoint. Within each trajectory, the two EMA copies start at the initial raw weights and update after every optimizer step:

$$\bar{\theta}_t^{(\beta)} = \beta \bar{\theta}_{t-1}^{(\beta)} + (1 - \beta) \theta_t, \quad \beta \in \{0.99, 0.999\}. \quad (2)$$

Neither EMA copy changes the raw training trajectory. There is no warmup, bias correction or sparse update rule.

The supplied residual denoiser has 296,450 parameters, 128-dimensional embeddings, width 256 and three residual blocks. Its lineage is the pinned AI-Scientist 2D diffusion template [8]. The study retained AdamW with weight decay 0.01, batch size 256, gradient-norm clipping at 0.5, and MPS float32 training. It used a fixed 100-step cosine diffusion-noise schedule and clipped predicted clean samples to $[-6, 6]$. Adaptations include the dataset generators, evaluation metric and constant every-step EMA; the study is not a reproduction of the upstream template’s reported results.

3.2 Pairing, evaluation and uncertainty

The independent unit is a training seed within a dataset, with $n = 4$. Confirmation seeds were 4001–4004 for moons and 5001–5004 for GMM8, separate from the pilot seeds. Each seed used 100,000 training draws, 2,048 held-out draws and 2,048 generated draws per model state. Initial weights, replacement minibatch indices, training noise, diffusion timesteps and evaluation randomness were matched across the relevant conditions. Actual arrays and prefix hashes were retained. The held-out and generation streams were separate from training.

The primary metric is empirical Sliced Wasserstein-1 (SW1), computed in float64 over 128 fixed projection directions with projection seed 7321. Lower values indicate better agreement with the held-out sample. For each seed, define the relative effect

$$\Delta_{p,D,\beta} = \text{SW1}(\text{EMA}_{p,D,\beta}) - \text{SW1}(\text{raw}_{p,D}). \quad (3)$$

Negative values favor EMA. Schedule contrasts are $\Delta_{\text{cosine},D,\beta} - \Delta_{\text{constant},D,\beta}$; duration contrasts compare 10k with 5k at a fixed policy. The interaction is the 10k schedule contrast minus its 5k counterpart. Absolute raw and EMA means are retained alongside these differences.

The analysis reports paired means, unadjusted two-sided 95% Student t intervals with three degrees of freedom, individual seed values, leave-one-seed-out means and exact two-sided sign-flip p -values. With four paired seeds, the minimum attainable sign-flip p -value is 0.125. Generated points, projections, retries and shared checkpoints are not additional independent replicates. The intervals are descriptive and imprecise at this sample size; no multiplicity-adjusted confirmatory claim is made.

GMM8 coverage counts a mode when at least 21 of the 2,048 generated draws lie within distance 0.45 of their nearest mixture center. Coverage is a secondary diagnostic and does not establish correct component masses or distributional equality. The frozen protocol and selected scientific inputs are supplied with the paper’s ancillary provenance.

4 Results

EMA-minus-raw SW1 was negative in 14 of 16 cell-level means, but only 2 unadjusted 95% seed-level intervals were wholly below zero. Figure 1 retains the original paired-effect figure. The complete cell

estimates and descriptive uncertainty are generated directly from the retained analysis in Table 1. The two intervals entirely below zero are the moons, constant-rate, 5k comparisons. The positive cell means at some completed cosine endpoints are small and have intervals that include zero; they do not establish harm or equality.

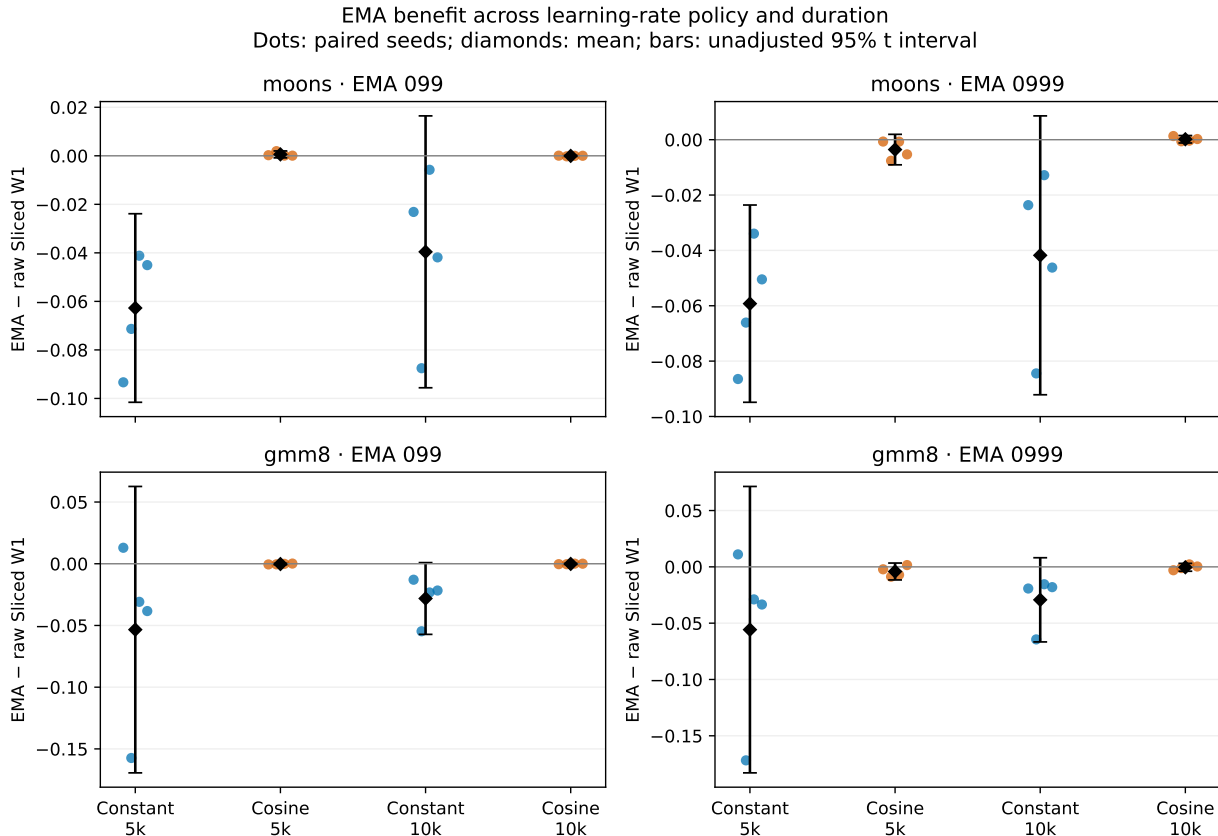


Figure 1: Exact figure retained from r07. Dots are the four paired seeds; diamonds and bars are means and unadjusted 95% Student t intervals. EMA 099 and EMA 0999 denote decays 0.99 and 0.999. Each panel has its own scale. Negative EMA-minus-raw SW1 favors averaging; zero-containing intervals do not establish equality.

Table 1: All paired EMA-minus-raw SW1 effects ($n = 4$ seeds per dataset). Neg. counts seeds with a negative effect. Exact sign-flip p-values are descriptive and unadjusted.

Dataset	Policy	Updates	EMA	Mean	95% t interval	Neg.	p
moons	constant	5000	0.99	-0.062730	[-0.101586, -0.023874]	4	0.125
moons	cosine	5000	0.99	0.000576	[-0.000852, 0.002004]	0	0.125
moons	constant	10000	0.99	-0.039576	[-0.095603, 0.016451]	4	0.125
moons	cosine	10000	0.99	-0.000041	[-0.000232, 0.000150]	2	0.750
moons	constant	5000	0.999	-0.059242	[-0.094880, -0.023604]	4	0.125
moons	cosine	5000	0.999	-0.003576	[-0.009083, 0.001932]	4	0.125
moons	constant	10000	0.999	-0.041781	[-0.092162, 0.008599]	4	0.125
moons	cosine	10000	0.999	0.000156	[-0.001190, 0.001501]	2	0.875

Dataset	Policy	Updates	EMA	Mean	95% t interval	Neg.	p
gmm8	constant	5000	0.99	-0.053400	[-0.169414, 0.062615]	3	0.250
gmm8	cosine	5000	0.99	-0.000242	[-0.000763, 0.000279]	3	0.375
gmm8	constant	10000	0.99	-0.028151	[-0.057218, 0.000915]	4	0.125
gmm8	cosine	10000	0.99	-0.000147	[-0.000531, 0.000237]	2	0.500
gmm8	constant	5000	0.999	-0.055769	[-0.182917, 0.071380]	3	0.250
gmm8	cosine	5000	0.999	-0.004116	[-0.011615, 0.003382]	3	0.250
gmm8	constant	10000	0.999	-0.029281	[-0.066649, 0.008087]	4	0.125
gmm8	cosine	10000	0.999	-0.000401	[-0.003890, 0.003088]	2	0.750

The cosine-minus-constant contrast in the paired EMA effect was positive in all 8 comparisons; 2 corresponding intervals were wholly above zero. Table 2 gives the prespecified schedule contrasts. The two intervals entirely above zero are again moons at 5k. The remaining schedule contrasts have wider uncertainty and include zero. The repeated positive direction is descriptive evidence within this design, not eight independent discoveries or a mechanism test.

Raw-weight mean SW1 was lower under cosine in all 4 dataset/duration comparisons. This distinction prevents a reduced relative EMA benefit from being mistaken for poorer absolute EMA performance. The absolute state-level estimates are retained in the ancillary analysis.

All 8 duration-contrast intervals and all 4 interaction intervals included zero. The duration and interaction estimates appear in the appendix. All their intervals contain zero, which leaves substantial uncertainty rather than demonstrating equivalent effects across durations.

All 48 retained GMM8 model states covered all eight modes under the prespecified threshold. The ceiling limits the diagnostic’s ability to distinguish these model states. SW1, component counts and inlier fractions remain separate retained measurements.

Table 2: Cosine-minus-constant contrasts in the relative EMA effect at each duration. Positive values indicate a smaller relative EMA benefit under cosine; intervals are unadjusted.

Dataset	EMA	Updates	Mean	95% t interval
moons	0.99	5000	0.063306	[0.023965, 0.102648]
moons	0.99	10000	0.039535	[-0.016330, 0.095401]
moons	0.999	5000	0.055666	[0.019748, 0.091585]
moons	0.999	10000	0.041937	[-0.007833, 0.091707]
gmm8	0.99	5000	0.053158	[-0.062767, 0.169083]
gmm8	0.99	10000	0.028004	[-0.000909, 0.056917]
gmm8	0.999	5000	0.051652	[-0.070912, 0.174216]
gmm8	0.999	10000	0.028880	[-0.007677, 0.065437]

5 Discussion

The descriptive pattern is consistent with earlier observations that averaging can provide a stronger relative benefit while a learning-rate policy maintains stochastic updates, and a smaller difference once the raw model has caught up during annealing [1, 6]. The newer benchmark already studies this interaction directly. The present results add a transparent small-model diffusion example with retained paired states; they do not establish a new general principle.

The experiment intervenes on the full learning-rate path. Changes in the terminal rate, the sequence of updates and the parameters entering the EMA occur together. A causal explanation restricted to the last rate would require another design. Likewise, the four-seed duration and interaction intervals do not support a reliable conclusion about how the schedule contrast changes with longer training.

Only two fixed EMA decays were tested. Architecture-dependent and training-dependent averaging profiles [4], uniform averaging [3], and switching averaged parameters back into training [5] are different choices. The retained data cannot rank them or justify a universal decay recommendation. The contrary ordinary-EMA outcomes in the literature also argue against treating an improvement as a condition for a valid research package.

6 Limitations

There are only four independent seeds per dataset, two synthetic distributions and one small denoiser. Confidence intervals are wide, multiple comparisons are numerous and generated samples do not increase the number of independent training runs. A zero-containing interval does not establish equivalence. SW1 on these two-dimensional datasets is not directly comparable with image-scale FID or classification accuracy.

The study does not test more EMA windows, SWA, post-hoc averaging, parameter-switching methods, different architectures or alternative optimizers. It keeps the diffusion-noise schedule fixed and does not separate the terminal optimizer rate from the rest of its schedule. Mode coverage reaches a ceiling and cannot rule out incorrect component masses. The finite-sample SW1 estimates are not debiased by subtracting a reference floor.

This publication revision reuses completed scientific outputs. Its targeted literature map is retrospective and not an exhaustive systematic review. The numerical checks in this revision recompute summaries from retained seed-level measurements; they do not rerun training or extract metrics anew from all saved samples. Earlier retained controller checks provide the separate saved-state replay evidence. The agent-assisted scientific critique and prose review are provisional and are not independent scientific peer review.

References

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A Duration and interaction contrasts

The following tables preserve every duration and interaction contrast in the relative EMA effect. Each estimate uses the same four paired seeds within its dataset. Intervals are unadjusted and descriptive. The source values, individual seed differences, sign-flip tests and leave-one-seed-out means are retained in the ancillary JSON files.

Table 3: 10k-minus-5k contrasts in the relative EMA effect at a fixed policy. Every unadjusted interval contains zero.

Dataset	EMA	Policy	Mean	95% t interval
moons	0.99	constant	0.023154	[-0.037175, 0.083483]
moons	0.99	cosine	-0.000617	[-0.002225, 0.000992]
moons	0.999	constant	0.017461	[-0.037112, 0.072034]
moons	0.999	cosine	0.003731	[-0.001185, 0.008647]
gmm8	0.99	constant	0.025248	[-0.061895, 0.112391]
gmm8	0.99	cosine	0.000095	[-0.000123, 0.000313]
gmm8	0.999	constant	0.026488	[-0.065668, 0.118644]
gmm8	0.999	cosine	0.003715	[-0.005093, 0.012524]

B Provenance, reference research and reproducibility

B.1 Scientific provenance and recomputation

The scientific records of completed run r07 remain unchanged. A selection manifest binds this revision’s protocol, figure and analysis copies to the original package index. This publication revision recomputed 114 statistical summaries from the retained seed-level measurements and matched the

Table 4: The schedule contrast at 10k minus the schedule contrast at 5k. Every unadjusted interval contains zero.

Dataset	EMA	Mean	95% t interval
moons	0.99	-0.023771	[-0.083068, 0.035526]
moons	0.999	-0.013730	[-0.072011, 0.044551]
gmm8	0.99	-0.025153	[-0.112347, 0.062040]
gmm8	0.999	-0.022772	[-0.110134, 0.064589]

frozen results. Checks cover per-seed contrasts, means, standard deviations, standard errors, intervals, exact sign-flip tests and leave-one-seed-out means. This is numerical verification of retained outputs, not a new experiment.

The full package retains unsuccessful attempts, the confirmation freeze, model states, evaluation streams and controller audits. Its public entry, <https://alohays.github.io/allagma/studies/ema-schedule/>, links the index, restoration instructions and audits. The ancillary selection manifest gives the exact source revision and per-file digests.

B.2 Reference research and local assets

The reference index covers seven primary papers and a pinned code snapshot. Bibliographic checks are separate from located support assessments. Competing findings, untested baselines and coverage limits inform this revision’s planning, implementation checks, analysis and writing. The notes explain why newer prior work limits its novelty.

Acquisition retains the seven PDFs and available TeX sources, three upstream repository files, one model artifact and one held-out/generation split from r07. They support reading and provenance inspection; no cited paper was newly replicated. Raw assets stay in a cache excluded through Git’s `info/exclude`. The paper archive includes notes and public-safe provenance without those downloads.

B.3 Attribution, disclosure and paper build

The owner supplied the author label *allagma* for this demonstration. Users configure authors per study; the framework does not infer them from Git metadata. OpenAI Codex generated or adapted the original scientific code and assisted with this manuscript and literature map. The adapted mechanism retains its AI Scientist Source Code License, included with the ancillary provenance, and AI-use disclosure.

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